

ActInf Livestream #024.2 ~ "An empirical evaluation of active inference in multi-armed bandits"

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all right hello everyone and welcome to actin flab live stream number 24.2 today is june 29th 2021 and we're looking forward to this follow up or jump off discussion with some of the authors as well as other lab participants on this awesome paper an empirical evaluation of active inference in multi-armed bandits welcome to the active inference lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find out more about us at the links on this slide this is recorded in an archived live stream so please provide us with feedback so that we can be improving on our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams it's great that there's so many on the call and i'm sure we'll have many questions and opportunities to raise our hand as well as for those in the live chat to ask questions whenever they feel like it we're closing out june with this dot 2 discussion on paper number 24. so last week we also spoke with sarah and dimitri and others that was very informative and now we're in the dot 2 we're able to take it to a few different places and raise up a few questions that were asked over the last week and the goal is really just to learn and discuss this paper and related topics we have a few cool related areas that we're going to be going into so we can just begin with a introduction round each person can say hello or check in say anything that they're thinking about today and then pass it to somebody who hasn't spoken yet so i'm daniel i'm a postdoc in california and i'll pass it first to blue hi i'm blue i am an independent research consultant based out of new mexico and i will pass it to ryan um yeah i am ryan smith i'm an investigator at the laureate institute for brain research um and i'll ask you sir hi i'm sarah i'm a postdoc at the technical university in prison and i'm looking forward to the discussions today and i pass on to dimitri hi everyone i'm dmitry markovic postdoc at technical university of raisin also uh chair of neuroimaging and i pass it to dave can he talk actually no my background's a basic psych and cybernetic learning thing and i'll pass this to stephen please hello i'm stephen i'm based in toronto i'm doing a practice-based phd around uh

social topographies and immersive experiences and i think i'll pass it back to daniel if i'm not mistaken yep great thanks for the cool intro round so sarah prepared this slide which we just want to start off with so go for it sarah how does this set the uh intention in the mood for what we're going to be going into today uh wait how can i make it so that i see it pick again no okay so um after the discussion last week and also the questions we've got we thought it would be a good idea to just make a short overview slide about what we understand as active inference and as cognitive neuroscientists you said in the discussion before also we are often or mainly interested in this action perception loop that an agent where we as living agents are in with our environment and that's what you see on the right in this loop and on the left side on the right you see the environment it may have hidden states that are the states that you may know from markov decision process processes in reinforcement learning but that may also be more abstract things that aren't observable tone agents like contexts volatility parameters anything that describes the environment and these hidden states may also evolve with time and then such an environment generates an observation that the agent then integrates together with its prior beliefs in order to infer the hidden states of the environment which is what we call perception then you can use this information to plan ahead and select an action accordingly and in turn the action may change the environment the agent is in and concretely how the agent side of this works in active inference is so that you specify a probabilistic generative model that contains the hidden states you are a priori knowledge or your assumptions about how these hidden states relate to each other which states may follow up on which states and so on it contains observation generation rules it contains actions or policies which in active inference are also treated as hidden variables that need to be inferred and additionally the generative model may contain its own parameters as a random variable so that they can be inferred and there was learned then you specify a probability distribution of approximate beliefs in theory you could also solve the beliefs of the generative model analytically but that becomes very computationally expensive really quickly or even analytically intractable so what you do instead is formulate these approximate beliefs which are supposed to be of a simpler form and for example in your approximate beliefs you could assume that all hidden variables are independent random variables which makes the beliefs very easy to calculate but may not properly capture the temporal dynamics for example of your environment and this this independence assumption is also called the mean field approximation but you could also say no with some hidden variables it's very important that they are allowed to co-vary and be dependent on each other and then you would have pairwise dependencies in these approximate

beliefs which is then called the beta approximation you may have different approximations for different variables some may vary together others are independent then you plug in the generative model and your approximate beliefs into the free energy and then you find your approximate beliefs at the minimum of this free energy and this way you can form beliefs about hidden states of the environment which really can be anything your location volatility of the environment the context you're in you infer beliefs about policies which is a probability distribution over actions from which you can then choose and you can even form beliefs about parameters of the model and there with update your knowledge of the model in each time step and then in the end you can use all this inferred knowledge to make actions um yeah we as cognitive neuroscientists are often interested in this whole loop but um what dimitri showed last week is also that in principle this can be really modular right so um in our paper for example perception was the same for all different agents whereas dimitri then plugged in different action selection algorithms one corresponding to the expected free energy and active inference but other well-known action selections algorithm too so um here you can really also play around with this framework and um use its modular nature you could on the other hand use the same action selection algorithm and compare different learning algorithms which is something we're also often interested about in our work and yeah with that that's it from me for now cool very interesting and anyone can raise their hand but i think the part that spoke to me there was just that it's a framework and so it's kind of like the framework of a desktop computer you can take out the memory or change out the hard drive and so we can talk about changing the environmental dynamics with respect to the hidden states and how they evolve through time we can talk about changing perception but we can also like lock in multiple of the components and then explore how different learning rules influence system behavior that's what we explored in this multi-arm bandit paper but it could be the same learning and policy selection rule but then a different perception rule and then those might map onto very subtle differences in um decision making of agents that we care about like what's the difference between somebody who has blurry vision but they know the language versus clear vision but then they're semantically unsure about what the symbols are representing um yeah stephen and then anyone else who raises their hand yeah this is really interesting and helpful to see your thinking here i'll just be curious how action states or um you know the idea of different types of states alongside these hidden states and just how that you treat those in your thinking um because there could be kind of the action states that are imagined um and the action states which are part of the kind of external environment that

you're somehow immersed in so in the end yes actions are also treated as hidden variables i think they are very separate from other from the hidden states for example because the space of actions determines what an agent can choose and you cannot choose a state u_m in terms of other hidden variables i think so for example if i think location versus context then u_m in my head the generative model becomes a hierarchical model exactly and in the end yeah that's also what i meant when i said you plug in your a priori knowledge or your assumptions about the environment because different types of hidden states will have different relationships to which and then you really end up with a completely different uh generative model just to follow up there sarah you said that when location and context are differentiated that it's a hierarchical model so what does that look like on layout you mean in this figure here like what's the setting where location and context are identical and so it's a non-hierarchical model versus a case where they are differentiated and then it is a hierarchical model so for example u_m if i want to model myself walking through my flat that wouldn't be a hierarchical model where just each location maybe the different room rooms count as states that i infer based upon the curtains i see for example but then if i want to introduce another person's flat then the transition dynamics may be different for their flat so for example in my flat i can go from the living room into the kitchen but in my friend's flat i cannot make the state transition and that's actually something i'm working with currently where an agent like the underlying decision process is very similar but an agent needs to infer which context he's in in order to then load the correct transition dynamics that reminds me of the spm textbook how it talks about hidden states being just the state that a process is in that generates outcomes so we have a clear distinction between the hidden stage which aren't directly observed but there is this hidden state like which flat am i in and then that is going to change the transition dynamics between the rooms but the observations are what are coming to the agent and then also there's like learning which we don't have here but that would reflect the prior beliefs being updated through time what about this planning stage like what is happening in this planning module u_m that's a good question actually nowadays u_m so this figures from my previous paper nowadays i u_m lamp in perception and planning together because it's really not that easy separable u_m the pro so in this case what i meant is perception is really inferring passed up into the current hidden states u_m to which i had the observations that belong to each state whereas planning is inferring future hidden states and future observations and potentially future rewards but in the end it's all one model and it's a chain to go through the states so the differentiation only makes medium sense i think it's rather one inference problem interesting

thanks blue and then anyone else so just thinking about metacognition and where that might fit into this model um i didn't actually have some um has published a paper on a meta cognition recently also so map control actually right uh but but yeah i mean one can just see uh one can stack multiple agents on top of each other and imagine as a kind of higher level agent controlling what the bottom level agent is doing and so this would be like well deep active inference models and in one of the recent papers we're also exploring this in a kind of as a meta control approach for describing uh like cognitive control so if there's some kind of downward causation does that then inhibit like what's going on at the like lower agent level yeah so basically right you can see kind of actions on the higher level uh are defining flyers on the lower level and selecting the policy space right so what kind of your level age lev lower level agent can actually do where this right this isn't now just kind of illustration of kind of separating maybe higher level cognition from the executive part which is kind of moving uh uh arms or muscles on your body uh it's it's interesting about that like kind of like a virtual machine or a hypervisor that there's this like emulated agent you know what would my better self do that's a classic metacognition question and then also it was just interesting that you know blue you asked about metacognition and then dmitry you write to meta control and so that really speaks to the way in which we're thinking about cognition as control and that's part of a control theoretic perspective which is that like the cognition planning as inference the cognition is action selection but then one piece that differs potentially is an active inference relative to just sort of the um way that control theory is often discussed is the insight of perceptual control theory which is that the planning as inference is being done in service of the expected observations and so we're planning and acting in order to control our perceptions and that's happening as part of this integrated loop so it's kind of cool that there's like this like good regulator theorem in terms of cybernetics and then it is like there actually is an emulated regulator agent and then um yeah pretty cool where does that play in with like art thought does that i don't know what does that relate to our actual experience of thought or does this happen all at the sub personal scale well i mean how i knew it it's really like a problem of separating different time skills and certain uncertainty associated with different time skills uh what you can do now and for example in the next couple of seconds is very different from what you can do in the next couple of months and potentially your long-term goals and plans have an impact on what you're doing currently uh and there needs to be a way how to resolve this uncertainty on different levels of representation and right from active inference what one's solution for that is just like you stack agents on top of each other which just representing increasingly longer

time skills interesting um so it and i mean as each agent i mean we can see it right as a little as a something which is separated from an environment through the marker blanket so basically through the actions and the observations it's making i mean you can also separate your brain in multiple nested blankets right where one part of the brain just informs and sends information to other parts and accepts signals as actions right so uh and i mean we know this from experimental research a lot of this is here are he constructed so all both in like temporal and like spatial hierarchies thanks dimitri stephen and then anyone yeah would i be correct in between perception and planning you've kind of got this more conscious belief awareness being sort of available and and the prior beliefs once you get from around the observations to perceptions might be below consciousness in the sense that it might be part of the visual system or whatever and i was just wondering how you see that transition from you know below awareness to phenomenological consciousness uh playing out with the types of models i know sometimes like ryan smith uses some semi-markovian processes which start with some um you know self-reports about beliefs so it's working at a level which is kind of building on top of the dynamics that might be going on in the kind of biology so i was just wondering how you see those different dynamics playing out and where you fit your models with some other models if they were sort of to play play with each other um if i may say so i think this relates back to dimitri's um statement about the hierarchy of time scales because um obviously if it's like my conscious thought is on a slower time scale for example than my visual perception is and i think the yeah my conscious thought is maybe rather responsible for the slower time where it helps me to narrate through the slow markov decision process whereas it would be not useful if i have critically think about any inference i make on my retinal image for example it's interesting you said there that the conscious thought it narrates because first off we've talked about narrative and active inference but also just like you described how there was this hidden state of which flat i'm in and then that sets the transition probability of the rooms maybe you have a narrative that sets up like which policies you're transitioning between like in baseball you know there's a story to the alternation of the innings and then it's like okay we're going to be enacting this policy because we're in this phase and so there's like a higher level narrative that helps connect when policy transitions might be um plausible to be adaptive and then that is almost by definition at a slower time scale than the play-by-play because the play-by-play is going to be a lot more like motor control the icicating and then as you slow down you get slower transition dynamics that are more and more narrative in nature because they narrate like phases of action but then the sub actions are very rapid i mean in a sense it

does have to do with a level of abstraction right if i see a leaf it's just a leaf it has a very concrete meaning in terms of my physical environment whereas my study program for example um is something a lot more abstract that is a lot harder to grasp without abstract states that you maybe need a narrative for more um ryan and then anyone else oh i mean i was just gonna say i mean you know like when we went through mine and chris's paper on consciousness a couple months ago or whatever i mean the this question about how conscious our conscious processes relate to active inference really any related kind of computational model is is a complicated one in part because what consciousness what the word consciousness is used to mean can be different in different contexts you know i don't personally think that um yeah i mean the question of what processes are conscious or unconscious in and of themselves doesn't really fall out directly from active inference right all we're doing is modeling things at different time scales and their relationships to one another um you know i mean in in the model of you know uh visual consciousness uh you know in that paper that we published a few months ago um you know the idea was very similar to what other people were saying you know which is this idea about different time scales um but but here i mean the idea the idea was something more like you have a particular level in a hierarchy that operates over a sufficiently long time scale that you can do the kind of goal-directed things that consciousness allows including generating verbal reports which themselves are very kind of extended um you know integrative policies um but that um you know but that also requires that it integrates information from enough different low-level sources right to do that um but beyond that you know talking about consciousness as itself the slower time scale process can also i think be a little bit um sort of subtle because what a lot of people mean by consciousness um you know has to do with the kind of subjective character of the kind of like moments moment you know phenomenological aspects of experience and that's clearly not happening on a slow time scale right like the those the moment-to-moment changes in conscious perceptual experience are fast um you know so what what we're doing in this kind of higher narrative level that i think you're talking about is something about you know integrating evidence for those over time and you know using them to come up with um you know certain sorts of longer time skill policies to adverse certain sorts of longer term skill policies um and you know so the way that you know we've talked about it previously is just that you know you have you can have these sort of depending on the precision of the interactions between the first and the second level uh in a model um if that precision is low then the lower level can um to a certain extent just kind of operate semi-autonomously in

which case the higher level doesn't really need to have all that much influence or know that much about right what's going on at the lower level um so you probably have as a kind of selection process where the precision gets turned up or down um at different times and for different hidden states um so the second level can kind of selectively become aware of and start integrating evidence with respect to certain lower level processes which simultaneously allows this kind of top-down control over those first level processes um you know so so with the perceptual phenomenology stuff it's probably more about um the moment-to-moment updates between lower level perceptual processes and these higher levels or timescale processes when the precision is sufficient for those updates to the second level to matter but anyway i mean that's more long-winded than i meant it to be but mainly just trying to point out that it's it's subtle and it means a lot and it depends a lot on what you mean by and it's not specific to active inference necessarily nice great i i think i mean the difficulty when talking about consciousness is i don't i mean we don't know what is what it is doing right what kind of problem is consciousness solving what would be like a hyper-intelligent uh organism without the conscious consciousness yeah i mean i think there are definitely there are definitely a number of ideas in the literature it's not you know it's not like it's completely pinned down right i mean like there are there are things for instance like requiring um working memory maintenance right like like extended working memory maintenance beyond a specific time scale looks like it can't happen unconsciously at least in current experiments um any kind of like multi-step um mental processes like for instance like um you can get unconscious priming effects where people can do something like two plus two um but you can't them but they can't unconsciously do two plus two minus four um right so anything that requires holding this kind of intermediate result in mind um to do a further operation on it is something that at least thus far nobody's able to do an experiment to um you know that people can do that unconsciously um there's um but you know what ryan would you think that consciousness is like a zero one state i mean from that respect right i mean right for example giulio tanoni right kind of sees this as a continuum like is really a continuum of states where you can just have higher levels and lower levels of consciousness well so so again i mean this gets back to what what you're using consciousness to mean right so in consciousness research there's a distinction between levels of and content of consciousness um now like content of consciousness is i think most people agree it's fairly binary i'm actually not totally sure what the um um what the iit crowd would say i'd have to kind of look back at um look back at some of the more recent stuff there but but um in terms of neuroscience you know it's it's it's very well characterized we're called

like ignition events that are that are non-linear all or none you know sorts of processes where if something becomes conscious you get this global non-linear just kind of broad uh activation widely across large-scale networks whereas if you don't get it you don't pass this like ignition threshold then you still get the local like perceptual like local activation in let's say visual cortex provision or whatever sensory cortex um but you don't get but it's just kind of linear and it doesn't kind of percolate up or pass this threshold to cause this large scale kind of more all or not kind of thing where that where the information becomes kind of broadly accessible throughout the rest of the system um so that that aspect i think has a fairly binary character to it um that's different than this kind of levels of consciousness and issue which is something kind of like the being in a predisposition to represent conscious contents um which would kind of be like a continuum from say like coma to like alert awareness um and that that probably just has to do with kind of like the state of cortical processing um allows for uh the the kind of dynamics that support this kind of all or none um the the stuff that allows for selective contents um but anyway i mean all this is going pretty far afield from from uh active inference in your guys's paper now so i don't want to uh go to the tractor delay it's fun stuff and it's important so um we'll go to a more applied question and then following this kind of round on an applied question we're going to go into a few code walk slash talk throughs as well as learn a little bit about a few approximations and some of other work by sarah so lars asked us a question on twitter anyone is always welcome to do and wrote i would love to hear any thoughts on how work on the multi-armed bandit might be related to real world problem spaces or applications how might active improvements in multi-armed bandit tasks translate to improving how some problems or decisions are currently solved we talked a little bit about this in dot zero and dot one but i'll go to any of the authors for a first take and then everyone else is welcome to give any thoughts or like ask a follow-up question so when you present the work and somebody just goes to this kind of obvious applied active inference question what is your thought well i mean as i said right multi-arm bandits are applied to a super wide range of real world so i opened a survey like a survey on practical applications of multi-arm bandits and contextual bandits one bone f and irish irina rich and jalal bonifo right uh so this is like one of the recent surveys i found 2019 on archive uh and they hear the list for example couple of domains where they are applied healthcare finance dynamic pricing recommender system maximi maximization dialog system telecommunications anomaly detection and i mean just by looking from that like i would first try things out with active inference based bandits in kind of non-stationary problems and this is from what i see dynamic pricing and your

commander system they are kind of in this domain quite obviously uh but one can also imagine that right uh telecommunication systems would also be a non-stationary problem and as kind searching for the fastest routing path and similar uh where different routes can change over time and then you have constantly ready to explore different channels for for passing along the information more optimally and right this is a kind uh practical uh uh domains where one could first try uh multi uh active inference based bandits um in as i study non-stationary kind of problems which is most of the other things i listed um this is a bit more difficult but first to see if there is a way to deal with um this kind of uh asymptotic bad asymptotic behavior basically over optimistic uh information search awesome sarah any thoughts on that or anyone else stephen yeah could you just repeat that last bit this you said something about isotropic behavior or some sort of type of behavior i didn't quite catch asymptotic right so this is like in in stationary bandits people are interested in the asymptotic behavior after like infinite many actions so to say uh and how the how different algorithms scale there whether they converge to a good solution or not and i mean this is what we find in uh right in the stationary problems active inferences uh too optimistic in a way it converges to a solution too fast as the way we define it at least and and that that's why it doesn't have good asymptotic behavior so if you would apply to non stationary problem when you would like with high probability to find good solution i mean active inference algorithm is not one to go for there again we have some ideas how to change this but i mean just based on expected free energy this is not something which behaves nice nicely so you almost have to do like a hack on it to keep bringing it back away from premature or yeah exactly it kind of just gets stuck prematurely into a into a solution which the agent believes it's a good one and the reason for this one can see immediately just from this from the term which drives the exploration which is this uh [Music] let's just expected information game right in reinforcement learning this is called like exploration bonus or something like this and this this doesn't increase with time so in a way uh all these previous algorithms like uh upper confidence bound ucb uh the reinforcement learning based algorithms they have a bound which increases with time so if you are not sampling from one arm uh this bound becomes bigger so your algorithm is kind of forced to switch at some point uh and this is not happening here so either one would need a different generality model or a different way to introduce more randomness into into the behavior one aspect of um it was uh figure one or no this figure figure two in the paper was that that uh on the top left there that the variance across active inference agents was increasing so it wasn't that like every instance was um slightly degrading in performance it was actually

that a small subset what you wrote there a small percentage of the ensemble did not find the accurate solution and were over confident in their estimate so it it does suggest a few ways in which maybe you know when the variance of a few parallel instances of active inference starts diverging that could be like a a warning sign that some of them are getting too confident too early and then also it's interesting how you explored um the learning rate um or lambda through time and just kind of said that okay there's no simple answer here but it's it's an area of future work for sure yeah i mean the problem with like parallel runs in practical applications you don't have that uh how do you say um that advantage is to keep in like simulations i can just run i mean any number of simulations just see how it behaves right in practice when you're solving the problem you just have one trajectory and then you have to kind of provide insurance that this sample or this sequence of actions will behave better than random or better and that there is some probability to converge over over long over long run and this is something which you don't get right with uh uh with active inference in stationary problems at least right in non-stationary we don't see this uh issue anymore because of the basically the generative model itself because the more you are the less you're exploring the more your uncertainty rises on the arms which you haven't observed uh because the agent believes that things will change over time simply this is also what helps it shift between arms and it's also very efficient to kind of extract information from them agent picks up really fast what what kind of arms were not simple and it's aligned with its beliefs i wonder if this also might reflect the difference between optimizing the model for practical application in i don't know a computer simulation program and the way that organisms behave you know it may be that you know we we prematurely shut off and let things become a habit but that may have downsides you know maybe that you know organisms just to minimize use of energy prematurely converge on something inaccurate and so maybe that's like gambling than that could it could show a fragility at times so there's maybe two ways that it's being applied you know yeah i mean i would say that organisms are never exposed to a stationary environment so right i mean in a way if you are not doing that you are suboptimal because you are living in environment it changes constantly uh right so in a way you can exploit this by now creating situations where uh well people behave weird and you have gambling issues and stuff but right um i don't think that's kind of disadvantage for well for what we evolved to do actually uh for that we are very good in doing finding good solutions reasonably fast yeah sorry i was just i was just curious i mean because you know i mean this is something that you know like with some of our like empirical work we've run into you know so like when trying to model uh like

change point detection tasks for example um using active inference where um yeah like same thing like the thing becomes too confident too quickly um but i wondered i mean because i mean in a lot of other empirical work like in neuroscience especially like it's it's pretty clear that people don't just kind of learn and then unlearn the like like reward probabilities or just whatever the kind environmental statistics are like when there's like abrupt changes right i mean what what people do is instead they infer that there's some new cause right there's some new light in context and then under that new context you basically just have really flat low you know really really small you know like magnitude concentration parameters and then you just build up right like your beliefs anew under that new context um and so there's there's like something so there's like that kind of approach either would require having some something hierarchical or having some kind of like additional hidden state factor that would correspond to context right and you know it seems like you could do it either of those ways um the um then the other thing to do would be to have some kind of like in the hp like in the hierarchical calcium filter where you have some kind of uh like dynamically adjusted learning rate you know or or really something more like um you know so in like recently when we were updating um some of this with our tutorial paper um you know like after talking with carl you know it seemed like it would be a good idea to also include this kind of like forgetting rate parameter as opposed to just like the standard learning rate parameter which is just kind of like a scalar parameter on the actual um on the actual concentration parameters right prior to prior to adding on the new counts um you know and so you could that's not dynamic but but at least it does um prevent you know it can act as kind of like a something like an implicit prior volatility that can prevent the thing from becoming too confident too quickly um but um but yeah i guess i just um um i wondered if you know you guys had like thought about or you know like played around at all with with something like that something like inferring inferring new kind of like latent context as opposed to just having to you know like the thing becoming too confident and uh having to spend a bunch of time over writing right it's uh you know it's old it's old beliefs which happens way too slow i mean in the end um my current model um is actually based on such a hierarchical model as you describe where my agent instead of unlearning all action outcome contingencies it um opens up a new context of have you inferred that the context changed and i think um when we were looking for proper learning algorithms for the paper we most definitely looked in contextual learning and um i think yeah in the stationary case there are no contexts um that's why we didn't introduce it there and um i think the uh smile algorithm we used for the non-stationary bandits also has a forgetting rate

um just like you say for getting right on the concentration parameters which i think is another reason why this algorithm worked well in the non-stationary case or better than in the stationary case and in theory i think we also tried some version of this where we have both contacts and getting parameters but yeah in the end then for the randomly moving bandits we had also thought about then having another layer to this hierarchical model that impersonal volatility to adjust the learn uh the floating rate and if you start that your learning model becomes very abstract pretty quickly maybe but overkill but i guess i guess what i don't completely understand still is so even in the stationary case if uh if you're getting rid of sufficiently high i mean that all that's going to happen is your the actual like magnitudes of your concentration parameters are never going to build up to too high of a value right so like does that not still help with the overconfidence issue i just would have thought it would uh well we don't have a per kind of forgetting parameter in non-stationary at least this shuts off because uh well assume agent believes it's in a stationary environment yeah i got a reasonable you could still put an agent in a stationary environment but it seems like it could be plausible that it has a kind of yeah right so yeah i agree i mean that's one way one could try out right to resolve the uh exploration problem mean stationary case right just simply uh giving agent to the wrong beliefs so things will continue just tell this we use the same learning the same bayesian belief updating algorithm for all the different action selection methods that we compared and then it's a weird interplay between the learning model that we chose for the stationary case and the action selection rule and i mean thompson sandling which uses the same learning does not have this issue yeah and that i think is also interesting so to the second part of the question where how would that translate to improving how some problems our decisions are currently solved i'm hearing a few things we talked about speeding up computation relative to other approaches but that's not a solved problem for example if there's a 10 times speed up but then you have to run 10 agents in parallel to get a good ensemble estimate then it's a wash so potentially for speeding up just the computational requirements for certain challenges a second would be that it might be possible to more rapidly lock in to dynamically changing regimes and to avoid some of the pathologies of model fitting in multi-armed bandit contexts and then a third way that it could translate to improvements would be it might reveal some hidden similarities between these different problems and settings like we already know that they're somewhat similar because we can apply a multi-arm bandit to you know health finance recommendation systems etc so we know that there's a lot of problems involving data that have similar enough structures such that a similar kind of general algorithm

can be applied but then it could be interesting once we have them on the common grounding of active inference to say actually you know the structure of the decision making is similar across these two settings or you know the telecommunications routing and the logistics routing are similar in this unexpected way so maybe insights relating to what kinds of tweaks an agent uh could improve on their performance with like what we're talking about here with the category categorical hidden states or learning and forgetting tweaks maybe some of those insights could be implemented in active inference and then more easily transferred across different domains so hope that conveyed some of our um uh thoughts on this question to lars and yeah yes i mean one issue there is just in different domains you will in principle have different generality models right i mean although the problem is the same multi-unbended i mean you would need different representation of the environment and this is then where the challenge comes potentially uh as if you can represent it as a multi-embedded problem you can uh choose different whatever action selection algorithm you find works best and in non-stationary situation at least from for what we investigated this seems to work well this doesn't mean necessarily that this generalizes one will still have to try out different things just to make sure but in the end the bigger challenge is like okay what is a good generative model for this dynamic problem which i have one can go there right with many different things so i mean for example what uh ryan also mentioned this kind of uh open-ended contextual learning i mean one can represent simple in in the environments where you don't know anything about what what's going on you can just do a non-parametric generative model like dirichlet process or gaussian process you just try to learn even what the model itself should be awesome so let's go to this little sub discussion on sarah one of your previous papers and then we're gonna turn to some notebooks and walkthroughs of the bandit project but hopefully this will be informative because first off belief propagation and message passing and these types of approximations are of interest to the lab in the community and also we're seeing a few faces that we can ascend active inference mountain on we have ryan with a matrix based matlab approach we will walk through in just a few minutes with a python based approach of the bandit and then this is a slightly different approach based upon the beth approximation so sarah anything you'd like to describe i'm sure this will be new to many people so it will be helpful to convey what you were working on here sure so in figure 2 the upper one what you see is the generative model of a normal observable markov decision process where the um in the upper row the unfilled circles are the hidden states sorry they were called as in the previous slide and below are the observations and then the whole um the dynamics of which states follow upon each other is

determined by the policy π on the left side and then what people often do is assume in this queue that we saw two slides before um the mean field approximation which means exactly which means that um for example here we have a bunch of hidden states h_t h_{t+1} and so on and then if you assume the mean field um the approximate belief distribution would be just $q(h_t)$ times $q(h_{t+1})$ and so on and so forth and there with you create an implicitly yeah you essentially treat all the as independent in your approximate beliefs and then their dependencies will be averaged out in figure 3 you actually see the inverted model and in with m you see the messages that are being passed in between nodes so if you it doesn't matter what approximation you use in the end you can calculate your beliefs with some sort of message passing except that now if you chose a mean field approximation and you estimate all hidden sets separately what we found is that actually they may not fit very well to each other so for when my agent predicted um how it will go through the grid under a certain policy it actually often predicted it will jump and go places that don't adhere to say transitions and there is of course leading to decreased goal reaching success and then instead of doing this mean fit we assume the beta approximation which instead of having $q(h_t)$ times $q(h_{t+1})$ and so on you have small pairs of joint distributions so $q(h_t)$ and h_{t+1} times $q(h_{t+1})$ and h_{t+2} and then doing the math you can show that if you assume this type of approximation and you plug it into the free energy you want to minimize the free energy the belief propagation method path passing algorithm comes out and so you can use the belief propagation and message passing under the beta approximation to calculate beliefs and this is actually exact on graphs with our loops and then you get a more appropriate joint representation or in this case of temporarily dependent hidden variables but i mean in the end yeah what i do nowadays that i also have a hierarchical model where now for example the parameters of the markov decision process are context dependent i look into my model and think which variables belong together and then i apply the mean the bit approximation in these parts but then some slower varying variables context i just use an infinite approximation because it it varies differently anyways thanks for that breakdown um what is message passing like who are the messages being passed between and does that reflect a variant on active inference or is it the same exact active inference model can be approximated or can be calculated through message passing or through other mechanisms uh correct me if i'm wrong dimitri but i think in the end all active influence agents do message passing depends on the approximation which type of message passing algorithm so um yeah another the better approximation it's belief propagation message passing i think there are

equivalences sum product algorithm also you just get different messages that may be better or worse but okay what are the messages essentially each node in the graph so each hidden variable sends other hidden variables that it's connected to a message about which state it should be in so ht would say hey we're currently here then i think in the next state we should be there and then vice versa uh ht plus one can send a message back that says hey we wanna be there next time sir where should we be now and so yeah these variables essentially send each other messages on what they should be so that they are in agreement with each other that's how you get essentially a probability distribution over what you think your which state you're in and will be in yet dimitri any thoughts on message passing or where do you see message passing fitting into bayesian statistics and a few other topics well i mean as sarah said mo i mean they would say all the algorithms are message passing so when you talk about mean field approximation this would be traditionally variational message passing it's called the algorithm right uh here uh this is like belief propagation it's a message passing algorithm based on uh marginal probabilities uh instead of in the variational message passing you have like uh expectations of the log of conditional probabilities right i mean this is these are kind of the differences uh what you're what you're getting and losing and i mean we have another paper with thomas parr and carl neuronal message passing using mean field beta and marginal approximations where we kind of contrast these different ways so who is interested in this topic can uh look into a bit unpacked discussion of similarities and differences in practice the difficulty with mean field approximation it's i mean for dynamical problems for numerical decision making it's not really a good approximation this is not something which one would use and implementation wise in inactive inference mean field approximation is not used on the dynamical level uh what what they're using in matlab for example is all this marginal approximation it's still kind of gradient-based method but it's computed slightly differently so uh yeah sorry uh well i just wanted to kind of wrap up is basically the more complex problem it is the more uncertainties you have on the state transitions the more difficult you would difficulties you will have it mean field and marginal approximation basically the bt approximation is the only thing which kind of uh corresponds to actually exact inference in the in non-cyclic graph so basically this is uh this is theoretical solution you know that you can be exact under specific conditions on the marginal um a yes i can only warmly recommend by adida and vice i think in 2001 called understanding belief propagation and its generalizations and i find it very didactic it hurt me a lot and they explain in detail how uh variational inference is also connected to message passing algorithms interesting so just to capture that

one thing you said there about the gradients the gradient it's sort of like your model is in a given spot on the landscape and then it checks the temperature and it goes in the direction of the gradient so we've talked a lot about gradient based methods with the straight line versus the iso contour message passing kind of breaks that down into a process so at each click of the messages are being passed back and forth to one another which is both computationally tractable uh it's also shown through some work of the bias lab uh burt devries and others that four specific categories of bayesian graphs that message passing algorithms are basically equivalent in the forney factor graphs the ffg which we're going to be learning about in the future and also this topologically puts you more into touch with the predictive processing for example the messages that neurons are passing to each other so it's one thing to say well it's as if the neurons are messaging to each other and that's doing a gradient descent but it's another thing to actually saying we have a message passing scheme for modeling how these message passing agents do inference so there's a few points of contact there that i think are pretty important and it's pretty it's also interesting that you brought up that early paper so it's we'll check that one out stephen did do you have a question i think it's been sort of covered really cool area though so thanks sarah for um sharing that just one last question on this before we go to the bandit codes and walk through like where do these approaches um are they converging and they're going to weave together more closely or is one of them like an umbrella over the other such that work will continue mainly under the generalized form where do these different approaches that we're talking about to implementing active inference where are they heading like is it is more development happening on the message passing approximation or on other modes of breaking down active inference i'm not sure i have the overview of hundreds of papers publish every month and that's active influence to say something like about uh as far as i mean our motivation is here we just want to have in different situation good enough inference approximations so in a way for me more important question is like what is a good representation of different tasks and environments to have rather than what is the best kind of inference algorithm to use uh because especially if i mean depending again on the environment you're working on and but in dynamic context it's uh difficult to get like lots of advantage with just improving slightly on the inference performance great just because there's lots of uncertainty in things and anyway change all the time so um yeah i mean with this we also kind of write for for this multi-unbanded paper we tested very lots of different algorithms uh there is kind of one of the notebooks in repository just kind of lists different things we tried out but in the end one doesn't see you i mean any reason why one algorithm would be

specifically a way is better than another it's just they're very similar uh subtle subtle differences great point that there's a lot of work on the comparability of different approximations and different algorithms but actually it might be more beneficial for a given application to focus more on how they're specifying the generative model and making sure that that really captures essential features of the because it's like okay let's just roll with active inference and spend our attention on the generative process and the generative rather than try to finesse potentially a a grossly inferior generative model with some better approximation there might be limited returns there so stephen and then we'll go to looking at some of the bandit code yeah this is a kind of general question related to that is i often think about sort of didactic deontique type sort of um ways of making inference um this sort of deductive thinking and such like and i'm wondering whether the message passing is is is more present in those types of models because it's it's something that's been detected in the environment and decisions based on that are being made and you've got then you've got your kind of inductive where you're trying to narrow the gap between a goal and then you've got your abductive where you're trying to build something up and infer from like a landscape that you're trying to work out what what is out there so to speak and i'm wondering if this is correct in my thinking that the message passing is more used when you have like a particular deductive reasoning approach and ad inductive or abducted ones would be different interesting question about how those different modes and types of logic are connected to message passing one thought which might be on or off base is that we're thinking a lot about how variables in a model are like nodes and then there's edges connecting the nodes that reflect the relationship between those variables message passing is just one way to describe as a model updates through time which information is being passed between the variables so it doesn't say anything about the mode of operation of an agent which might be engaging in different kinds of logics and i think that's in really excellent question like how how do we um break out of the known with respect to how our algorithms update and it does actually touch upon this mean field approximation for example if you think that all through all time past present future that there's some stationarity of the hidden state then the mean field approximation will but then if there's going to be a change the state then the mean field is potentially going to give a misleading outcome but in any case message passing is just describing sort of the mechanics of how the model updates and which information between variables is connected but importantly which variables are not connected like the observation at a time point doesn't influence the observation at a time point directly but it could via a specified path of message passing let's look a little bit at the bandit

code um so the the links to the github dye markov you have the right name to work in the area um so we have a few of these notebooks up and uh is there we can look at the overall notebooks folder or do you have a sense of which of the notebooks might be interesting to walk the record go to a few or if you want to jump to a first one i think just this first one think about this maybe oh go ahead uh yeah there are a couple of things i mean a couple of notebooks which are not immediately relevant for the paper so which we can focus just on the things which are part of the paper or i can just generally kind of talk also about these other things which are just process of thinking about the problem okay how about before we even how do you as a researcher working on this area keep that separate like the paper specific developments but then your overall developments do you find that you're on an overall development mood and then you dip into specifying a paper or do you pursue the paper and find that you have more general insights while you're working through the problems well i mean i pursued the paper but i mean then i mean there are lots of branching pads on that way in a way you have to figure out uh what's potentially interesting what's relevant and so part of this code is just exploration a bit of things topics which were interesting for me but which turned out not to be so important in the end just maybe for some other paper uh and things which are just focusing exactly on the comparison of multi-arm-banded algorithms and right uh discussing this part uh so in a way it's difficult to combine lots of potentially unrelated things into one paper so one always has to make some favorites in the end i knew it would be a both type of question because it's something that researchers are often you know interested in general questions but we need to deliver on specific research projects with a defined scope and conclusions as well so it's just cool to see that this repository holds a little bit of both so so for example this uh notebook which you opened first like expected free energy comparison this was just my contemplation of just different ways you can define expected free energy so typically people will think about expected free energy in this terms of expectations of their outcomes but for example another question is okay but why not why not computing in terms of expectations over states and there is the relation between these two right uh one is an upper bound on another so basically you see this last relation there is s of π there is g of π and there is i of π uh and um basically g of π would be expected free energy in terms of expectations over latent states so this first and this is upper bound on what this would be like expected surprise for me uh in equivalence to the free energy being the bound on marginal surprise like log likelihood or surprising uh and then there is the i of π which is just kl divergence between posterior prior which gives you then something else and anyone can also think well we can

select policies or make decision-making algorithm based on any of on these quantities and what happens when you use one or over another so this is something which uh i was just testing out for myself uh and i'm still not clear what to think about this so that's why i don't have a paper it this is very interesting um it's like it's all conditioned on policy with π and then we're approaching it from the top and from the bottom and so the free energy is like being sandwiched in between these other approximations um and then you wrote here that the minima of i and g of π match but s is giving a different minima so what what was curious about dot co no at least in this example right uh so i just kind of build up simple example and one can see that optimal policy uh or like minima of these different quantities is different and one can also think probably of different examples well this will i mean this relation will not hold anything but what i just wrote as a comment there uh but i mean this is more like than practical question so if you would then build an agent uh which one of these quantities should you use they are all kind of effectively can be seen as a right expat expectation of a future surprise uh and having different bounds on that uh expectation as an approximate quantity uh so just the question is uh and i also found in different uh problems depending how i formulate one of these algorithms works well not algorithms but objective functions let's call them works better so so in this particular multi-methods task which we if we explore the g of π is not so what should i mean what should corresponding to expected free energy this is not uh behaving that well so in a way you don't get uh such a good performance but i can slightly change the task uh and i i can get better performance with geophy or sf5 so as i said i'm not sure what to think about this still so cool well it's really like exploration different things we'll we'll have you back on this topic when you're in a different phase do you want to look at inference algorithms comparison or is or well yeah so inference algorithms as i said this notebook and just list some of the things which we consider right in the literature when you think about this um well approximate difference problems in changing environments one can think of different ways how to solve this uh what we used here is the radius representation which comes from change point models as an approximation for the task which is a good approximation for this kind of kind of pitching bandits which is what we uh what you show our job showing here right for example this is the question blue had last week also so this would be how probability changes in us in switching bandits on one arm uh over time right now probability of generating uh one or let's call this a reward uh this is what this plot is showing uh and right and this is the switching concept that after each uh whenever switch occurs we are just sampling the new probability for each arm so this is this kind of uh setting non-stationary uh

uh difficulty right so that basically this difference between the best arm and second best is it varies over time uh and then you have like a drifting dynamic for which just like the different generative approaches uh model would also be better uh but then one can also say well we can use any generative model for any of these problems and let's just see which does inference better are there any differences there right so can you just uh use different representations in for different underlying uh environmental dynamics in a way miss specified in the model but still doing reasonably well in for the inference part and right when you kind of look at the results kind of posterior expectations you get over time uh different approaches to lead to very diff similar results in the end so there is no kind of strong advantage or disadvantage of one or another and that's that's the reason why we just picked the simplest thing which can the more most efficient basically algorithm because then it's much easier to scale to more arms more time steps okay very interesting this is a pretty thorough walk through the hierarchical variational smile yeah exactly it just describes right the generative some of the steps one needs to take to get to the posteriors uh they're also uh then just implementation of the algorithm is unpacked there but beside this right what we used in the paper we also implemented some of other approaches which are non-variational kind of asian uh which is seems to be quite good and i mean right on average it performs better it's like more optimal representation now this comes from i think also recent paper and multi-armed um do you remember sarah they'll first start a louis at all maybe or unfortunately written paper and one can see that it does slightly better job so that would be i guess this algorithm you're showing now yeah what are the lines representing the red blue green and then the sort of flat the green is just for the ex posterior expectation or like reward probability so right the perfect algorithm should just match green with blue and the red is basically a change point inference so basic posterior probability that the change occurred at this specific uh moment of time uh and as you can see this process is quite noisy in a way right you you have lots of small errors in a way or slight jumps in in places where change didn't necessarily occur you know what what this reminds me of is um the blue is some hidden true price of an asset and then the green is like the markets tracking that price or value and then the red are like orders like buy and sell orders on the market that represents the underlying situation changing and it's like just uh maybe well i mean markets yes ma'am i can see markets is doing some kind of difference on the true friend by value of the price uh right this will be kind of distributed inference problem yep i mean it's the whole no one knows the price of a pencil but maybe the person making the eraser knows when the price of rubber changes a little bit but but their colors are just the coincidence okay and then here we

see something um yes here is this this is a if i remember correctly this would be the hierarchical ocean filter applied to to the same so this is work from matisse christmas so so he actually in his first paper he he did apply to inferring the price of an asset over uh so when he introduced that but but found lots of uh applications in the cognitive neuroscience and just uh understanding how people adjust to volatile environments so there so what we have here is that this kind of change probability is constant over time but you can also think of the environment the change probability itself changes over time so that you you need to kind of adjust uh to these changes also uh so that would be also quite one uh straightforward extension of this multi-arm banding problem like different types of dynamics and uh testing out then more complex generality models and approximations to deal with that problem and so right we are just comparing now how this algorithm tracks the underlying price value in different environments so i i have a question about um what action does so here the underlying generative process is stochastic but the actions don't change the process like choosing a different slot machine isn't changing the probability of slot machines so it's almost like a little bit more of a niche modification setting or can it just um be directly put into the model that certain actions actually change aspects of the underlying generative process or is that kind of a feedback between action and then future hidden states is that like another module that has to be constructed in well for what we have implemented here that would definitely need an extension so we kind of this is more than this general representation which of uh also implementation of active inference which for example is implemented in spm which then helps you deal with uh these more extended problems so here we really are working with the uh simplest algorithm just for having it like very efficient and compact so that it can scale easily uh what the more general you are and trying to capture many different problems the more difficult to have with scaling right uh it's it's like inference all the way up or all the way down and it's it's a theme that we'll return to many times which is that it's all good to track the absolute value of what you're interested in but then the uncertainty on that and then the uncertainty on your uncertainty about that can get you into this infinite recursion so you just do you go quick and dirty and just have a simple idea of how variance and higher order uncertainties propagate or does one fully specify all the possible ways that uncertainty can exist across multiple levels which can get to an explosion of the computational requirements really fast well i don't think that necessarily hierarchy is the issue but it's more right if you then start assuming that your actions change the states state changes uh or transition matrices in this stuff right so basically you're effective with actions you're modifying the state transition

matrices then this requires that i mean you also think about the planning problem and it's not any more simple um action selection problem but it's then becomes a planning problem uh and this makes things more complicated if you introduce such an environment because then it depends where you are at different moments of time in different states yes because we've seen in the markov decision processes that policy π plugs into b which is the transition matrix between and so that's like actions changing the way in which hidden states are inferred to change whereas the one-step decision making is um doesn't have to be done in that same uh way but yeah and i mean i think also i mean at least original implementation in spm what is used this doesn't scale also very well for this type of problems and different groups or people have started exploring like monte carlo research and other methods which actually allow you to then figure out potentially best policy in a very complex high dimensional problems all right but but i mean as long as you're kind of in domain of cognitive behavior in neuroscience you can like get away with this by just making your task reasonably simple you have this control so yes but even as as ryan like pointed out earlier that uh real humans even when you control the experiment or you think that you're introducing like a gradual change in a parameter they might actually be cognitively doing a different type of inference yeah humans are problematic i don't like them oh humans we should do experiments with robots well maybe that will be the uh i mean we brought up the conversation of course logistics planning motor behavior exploration exploitation spatially those are things where maybe having a defined digital twin for some robotics and then we go from in silico to to starting to introduce the element of the human and the unknown um stephen sorry i think i mean like but in in behavioral experiments you always have this problem of uh just like convincing yourself that the model you're using is something which reasonably well represents with humans are doing and you can be quite certain that this is not what they're doing exactly right this is just another approximation of all the complexity which we kind of uh have in well our non-parametric representation of the world uh right and i mean kind of there's this issue right you can kind of get people to perform very well on the task through lots of training but then it's like okay this really what i want to kind of test experimentally whether like how can humans learn to do this task well or i'm actually interested what people are doing when they are solving any tasks how they represent the environment date the time representation how is this incorporated in their decision uh model so and i i think it's i don't know how other people feel maybe ryan i can comment on this but for me it's always like a difficulty to to deal with it's like it always certainly or whether you're doing the right thing like simply enforcing something on some task

on people yes thank you stephen yeah so talking about this the problem with humans but um is i'm interested in how the precision parameter you talk about the precision parameter and it helps to sort of determine whether someone's going to explore or exploit the context i'm interested in how exploration can become a pragmatic uh this precision about the usefulness of something that's exploratory i.e what art or an experience of some sort and how that meaningfulness in the future could offset a pragmatic gain in the near term so i i i'm interested in this the way that precision fits in with that um and sort of the evolution of that and they even talk about that a bit with the charge with casper hess by affective charge is how your precision about an expectation has been violated or not you know it's not necessarily whether it's good or bad it's whether your your prediction of how well you could expect something to happen suddenly got violated and that amplifies everything um so i'm interested in that because i'm trying to create immersive experiences theater and places like that but i was just wondering what your thoughts about the precis how that precision parameter fits in with that dynamic and how that could be extended or if there's other other parameters that kind of can fit in there as well um i'm not quite sure that i understand the questions but let's see the are you asking like can is precision always relevant yeah can you stack the precision with the pragmatic if that makes sense so say you've got low precision but you have a high precision over the fact that exploring the low precision would be useful so the two things kind of stack on top of each other as being a sort of a pragmatic epistemic game i can see that being used yes so i mean we in this other paper on meta control we are playing with this a bit differently so we are saying that uh you can kind of control your exploration tendencies if you learn over time the exploration is bad and this is kind of where this kind of stacking of different levels of the hierarchy here like a higher level kind of agent controlling the lower level comes in play all right because simply if this higher level observes that over time uh you're like the agent is not performing well in a way it's not reaching the goals efficiently then it kind of punishes exploration and learns uh just to be more exploitative in this specific settings uh and the other way around where we can make kind of a setting where the exploration is beneficial always and uh right yeah and this is then the agent learns to kind of behave in this way better so one can like then i mean assume in real situations this is something which people learn right so just if they observe that depending on what they do they gain more or less they will also adjust their uh tendencies uh and associate this with different contexts right so you could have almost like it's not just precision you could have like another generative model managing those yeah exactly i mean you can just have another kind of higher level representation which controls uh

priors or precisions on the lower level so right and basically inducing different behaviors through to connect that to the um ostensive cues it's almost like if you had a a sound or a cue that said okay now we're in a brainstorming period okay now we're gonna drill down and by alternating between a brainstorming period and by drilling down we're going to have the right kind of outcomes but giving a cue for when one should be in a more exploratory mode versus potentially like more working with what is already known but this looks like an interesting paper this meta control of exploration exploitation dilemma and also framing it in terms of the hierarchy of time scales whereas often it's framed in terms of um i guess just instantaneously what maneuver would be best whether one is instantly preferring exploration or exploitation rather than uh through deep time the interaction with reviewers during that paper is what motivated these papers so that's that's like sticking like okay i really need something to just explain uh unpack some of these similarities and differences between active inference and everything else around nice um well if there's any other sort of closing thoughts this has been an awesome set of discussions i think we'll have a lot to think about and hopefully people will work through these notebooks keep an eye out for when the final paper is uh published with so you got this one fully published while the multi-armed abandoned is is still not ready it's still under review yeah we are finished yeah hopefully soon cool it's a first revision around currently well to anyone who's on here any other thoughts or questions or what do we take moving forward i'm just thinking about different tabs i having a regime of attention on as the multi-arm bandit you know do i check something i haven't checked in a while because i'm uncertain or should i pull back to my higher level and be like you know what it doesn't even matter if i'm uncertain about that tab i should just stick on this one i'm sure there'll be some fun um yeah dimitri well i mean it's a very general problem as i said so most of the problems are multi are invented it's not surprising right awesome well sarah and dimitri thanks so much it's great to have your like engagement from the 1.0 on through it made it awesome for the lab so thanks everyone for watching and until next time thank you for having us here yeah thank you yep yeah thanks very much until next time bye bye